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Determination of Replacement Cable Force of in-Service Suspension Bridge Based on Refined Modelling Approach

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Abstract. In this paper, the replacement cable force of suspension bridge slings in service is systematically analyzed by adopting the coupling analysis model of internal force influence matrix and displacement influence matrix. The sling replacement project of an existing bridge serves as the basis for considering the technical condition and stress state of slings and other components comprehensively. Key control indexes, such as internal force, offset, linear or adjacent cable force change, are defined according to vehicle traffic requirements and structural safety considerations. Refined finite element numerical calculation is employed to determine the target cable force applied to the replacement sling, while analyzing changes in key control indexes of relevant components and structures during adjustment. The target cable force of the replacement sling is adjusted and determined based on a comparison with the target demand index. Compared with the traditional influence matrix method, the double matrix method also has the advantage that it does not need to select the initial adjustment cable force, which avoids the problem that the convergence result may not be obtained, and aims to provide a basis for the subsequent calculation of cable force replacement when the suspension bridge is in service.

Keywords. Sling replacement of suspension bridge; In-service bridges; Double matrix approach; Sling Force calculation; Bridge alignment effect

1. Introduction

In order to enhance the intrinsic safety, service durability, and reliability of long-span suspension bridge structures, as well as mitigate or prevent the risks [1-6] associated with safety accidents caused by the deterioration and fracture of suspension cables during operation, the replacement of suspension cables based on structural requirements has emerged as a crucial measure [7] in ensuring the safety of suspension bridge structures. The replacement of suspension cables should aim to restore the suspension bridge to its ideal state of completion. The calculation of replacement cable force necessitates comprehensive consideration of not only the changes in overall structure and traffic load on the bridge but also the time-varying effects of material properties, such as the impacts of concrete shrinkage, creep, steel structure corrosion, and aging of components, on the overall performance of the bridge. Given that cable force serves as the pivotal factor governing the shape of suspension bridge cables, current research on suspension bridge cable forces predominantly revolves around two theoretical frameworks: the cable mechanics-based analytical method and the finite displacement theory-based finite element method.

The sectional catenary method is extensively utilized in analytical approaches, as it deduces linear parameters [8-10] based on mechanical equilibrium and geometric compatibility conditions, eliminating the need for matrix iteration and complex algorithms. The cable saddle shape and tangent point position



are pivotal factors governing cable force and the main cable profile. By establishing the relationship between the tangent point of the cable saddle and the main cable, the analytical method directly computes the tangent point's location during construction and upon bridge completion. Currently, the analytical calculation of suspension bridges primarily focuses on the line shape and force of the main cable. When considering the influence of the stiffening beam and bridge tower, the connection between the main cable and other structures is typically achieved through rigid connections of the suspension cables, neglecting the coupling analysis between the main cable and the main bridge body. Consequently, further advancements are required in the analytical calculation of suspension bridges to enable the coupling analysis of various main structures, thereby enhancing the overall accuracy of computations [11-14]. In contrast, the finite element method employed in suspension bridge analysis discretizes the structure into finer units, utilizing different boundary conditions, and enables the calculation and analysis of various bridge structure couplings and different stress states [15].

The theoretical framework and methodology employed in the calculation of suspension cable replacement significantly impact the structural safety during the cable replacement process. Moreover, the replacement of the cables themselves can contribute to improving the overall shape and internal forces within the structure. In the development of replacement plans, careful consideration is given to the influence of structural deterioration on bridge performance, ensuring that the bridge achieves the objective [16] of maintaining a safe and optimal stress distribution.

This study focuses on the technical condition and stress state of suspension bridges, as well as other components in service, while considering the disparities between the original design or bridge condition, traffic operational conditions, and changes in environmental climate conditions. With the aim of meeting both vehicle traffic requirements and structural safety, the key control index for the suspension bridge structure, which pertains to changes in cable forces (linear or adjacent), is firstly defined. By taking into account the alterations in service conditions and boundary conditions following the replacement of suspension cables, along with the desired key control indicators, the replacement cable force is preliminarily determined. Refined finite element numerical calculations are then employed to assess the impact of replacing the sling cable force on the static and dynamic performance of relevant components or structures, as well as the key control indicators. These results are compared with the quantitative indicators reflecting the target demands, and subsequently adjusted to determine the final replacement sling target cable force.

2. Cable force adjustment scheme

The influence matrix method takes into account the combined effects of two types of influence matrices: the displacement influence matrix and the internal force influence matrix. To consider the double influence matrix, the influence matrix of cable force replacement on the internal force of the suspension cable is defined as $[C^i]$ and the influence matrix of cable force replacement on the control node of the main structure of the bridge as $[C^d]$. In the internal force influence matrix $[C_{ij}^{ff}]$, the element C represents the variable that causes the internal force of the sling in section i when a unit replacement cable force is applied to the sling in section j . In the displacement influence matrix $[C_{ij}^{dd}]$, the element C represents the variable that causes the displacement of node i when a unit replacement cable force is applied to the cable in section j . The construction process of the internal force influence matrix and the displacement influence matrix is as follows:

1) Considering the structural characteristics and load conditions of the bridge, calculate the internal force vector $\{T_0\}$ of the sling in the initial state and the displacement vector $\{D_0\}$ of the control node of the bridge structure. The control node can select the lifting point of the sling on the main cable and the corresponding node at the saddle position of the main cable.

2) The unit replacement cable force is applied to section i , and the internal force vector $\{T_i\}$ and the displacement vector $\{D_i\}$ of each control node under the joint action of the initial state and the unit replacement cable force of section i are calculated. If the unit adjusted cable force is not enough to cause

the change of internal force or displacement of the structure, or the result difference is reflected in no clear physical significance of the exact number of significant digits, the adjusted cable force can be appropriately amplified Δ times, calculate the internal force vector $\{T_i\}$ and the displacement vector $\{D_i\}$ of key nodes under the joint action of the initial load and adjusted cable force Δ ;

3) According to the principle of superposition, get the i^{th} segment sling unit delta change the hanger forces under the joint action of the sling internal force change vector $\{C_i^t\}$ by $\{C_i^t\} = (\{T_i\} - \{T_0\}) \Delta^{-1}$, and get the i^{th} segment sling unit delta change the hanger forces under the joint action control node displacement vector $\{C_i^d\}$ change by $\{C_i^d\} = (\{D_i\} - \{D_0\}) \Delta^{-1}$;

4) The internal force influence matrix $[C^t]$ and displacement influence matrix $[C^d]$ of the whole sling system are obtained by performing 2) and 3) steps for all slings.

According to the above steps, the influence matrix of the sling force on the internal force of the cable and the influence matrix of the displacement of the node can be determined, and its flow chart is summarized as Figure 1.

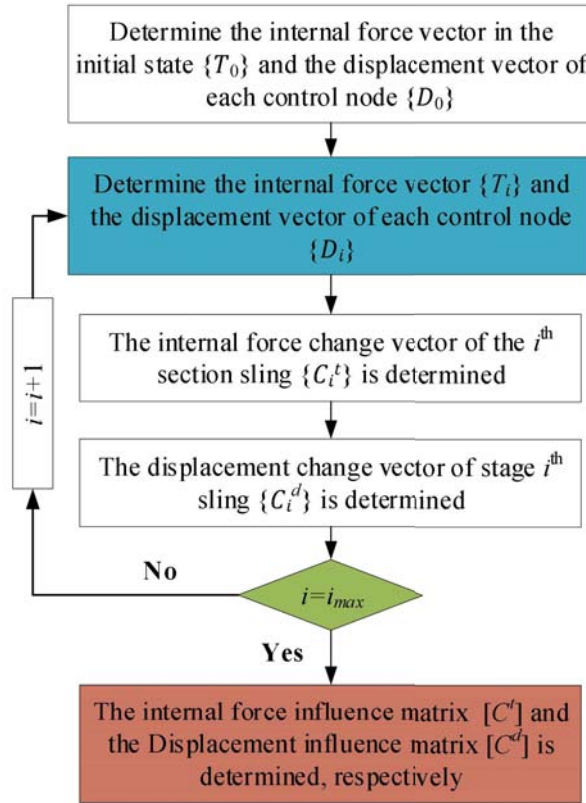


Figure 1. Flowchart of determining the influence matrix of cable force.

After determining the above influence matrix, the displacement change vector of the control node $\{D\}$ and the internal force change vector of the sling $\{T\}$ can be determined by equations (1) and (2) :

$$[C^d] \{X\} = \{D\} \quad (1)$$

$$[C^t] \{X\} = \{T\} \quad (2)$$

Combined vertical (1) and formula (2) yields:

$$[C^d][C^t]^{-1}\{T\} = \{D\} \quad (3)$$

Assuming that the target cable force vector is $\{T_i\}$ and the initial cable force vector is $\{T_0\}$, then we have $\{T_i\} = \{T_0\} + \{T\}$. Similarly, the relationship between the target displacement vector and the initial displacement vector is $\{D_i\} = \{D_0\} + \{D\}$. Then equation (3) can be written as:

$$\{T_i\} = \{T_0\} + [C^t][C^d]^{-1}(\{D_i\} - \{D_0\}) \quad (4)$$

Formula (4) gives a theoretical method to invert the target cable force vector $\{T_i\}$ by solving the cable force influence matrix under the target displacement state $\{D_i\}$.

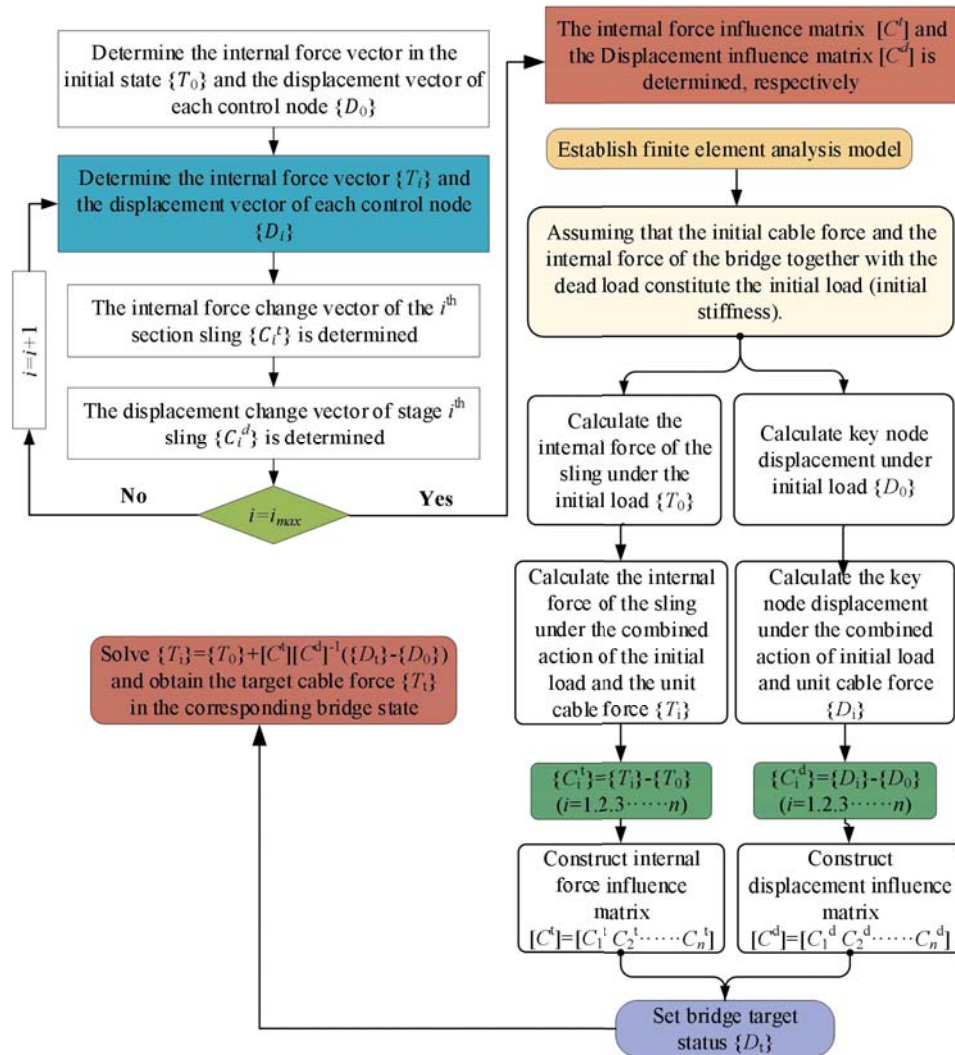


Figure 2. Process of bridge cable force calculated by double matrix method.

The calculation process of the bridge cable force determined by the double matrix approach is shown in Figure 2. As one of the sources of the geometric nonlinear effect of suspension bridges, the initial internal force effect of the structure is determined by the structural characteristics, boundary conditions and initial load of the bridge. The initial stiffness of the structure is formed by assuming the initial cable force and the internal force of the stiffened beam according to the internal force balance method.

3. Bridge case study

3.1. Bridge Project Overview

The Bridge A is a self-anchored cross-sea suspension bridge spanning the sea, which is located in Guangdong Province, China. The main bridge consists of three spans measuring 302.0+888+348.5m, with a width of 32m. The side cables are straight back cables, and the bridge deck accommodates two-way six lanes traffic flow. It comprises six key components: stiffened steel box girder, main cable, suspenders, towers, foundations, and anchors. A total of 72 groups of suspenders are installed on the suspension bridge structure as illustrated in Figure 3.

The full-bridge simulation analysis of Bridge A was conducted using Midas Civil 2021 finite element software, and the model schematic is shown in the Figure 4. Timoshenko beam units were employed for cables and suspenders, except for cable units. The main cable is constructed with prefabricated parallel strands, consisting of 110 strands in total. Each strand comprises 127 galvanized high-strength steel wires with a diameter of 5.2mm, exhibiting an elastic modulus of 196GPa. Simulation was performed using the equivalent area method.

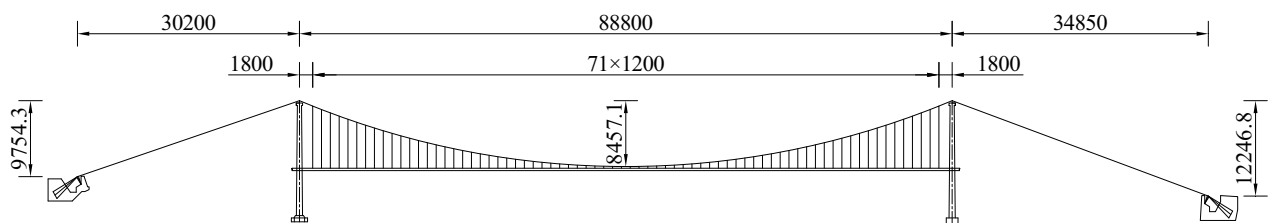


Figure 3. Structural layout diagram of Suspension Bridge of the Bridge A (Unit: cm).

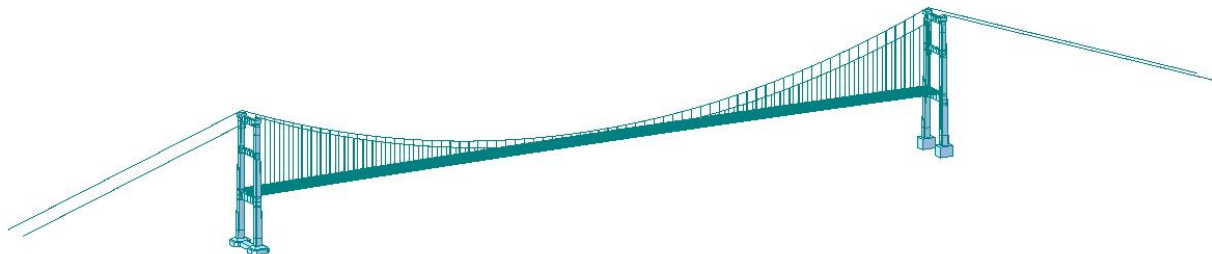


Figure 4. Three-dimensional analysis model of suspension Bridge A.

3.2. Target Cable Force Calculation of Suspension Bridge Case

According to the condition of the primary bridge, the influence matrix of the entire bridge is extracted. Due to the symmetrical left and right amplitudes and pile sizes, only half of the bridge is analyzed. Therefore, the dimension of the cable force influence matrix is 36×36. Based on simulation analysis results, we can obtain the initial cable force vector $\{T_0\}$ and initial displacement vector $\{D_0\}$ under the primary bridge. The rational state of a self-anchored suspension bridge includes internal force and linear control factors. The internal force control factor consists of stress states in main beams, cable towers, main cables, and suspenders. The line shape control factor includes main beam line shape and main cable geometric line shape. Under target bridge conditions, replacement cable forces should maintain a downward deflection in accordance with specified line shapes for main beams at points where slings intersect stiffened, i.e. $\Delta_i = f(x_i)$, ($i=1,2,3...n$), totaling 36 components.

Apply F_i replacement cable force to the i^{th} sling unit. After the corresponding construction process is added, the finite element result is $\{T_i\}$, which is different and normalized to obtain the change vector $\{C_i^f\} = (\{T\} - \{T_0^i\})/F_i$ of the i^{th} unit cable force. This vector is the i^{th} column of the internal force influence matrix of the suspension system. The other columns of the internal force influence matrix of the suspension system are obtained by using the same calculation for the remaining cables. The internal force influence matrix of full rank 36×36 is finally formed. Similarly, replace the cable force vector with the displacement vector and repeat the above calculation process to obtain the displacement change vector $\{C_i^d\} = (\{D\} - \{D_0^i\})/f(x_i)$ of the first sling unit cable force adjustment, which is the i^{th} column of the displacement influence matrix of the sling system. The other columns of the displacement influence matrix of the sling system are obtained by using the same calculation for the remaining cables. Finally, the displacement influence matrix of 36×36 full rank is formed. According to the target bridge state, target displacement vector $\{D_i\}_{1 \times 36}$, $\{D_i\}$, $\{T_0\}$, $[C]$, $[C_i]$ can be obtained by substituting $\{D_0\}$, $\{d\}$, $\{T_i\}$, $[C_d]$, $[C]$ into equation (4).

The bridge cable force under the target state and bridge design state are illustrated and compared in Figure 5. It can be observed that there exists a significant disparity in the ratio of cable adjustment among each boom, with adjacent nodes exhibiting opposite amplitudes of cable adjustment. This indicates that an unreasonable predetermined target linear shape results in excessively large differences in cable forces between adjacent cable poles, leading to substantial nonlinear fluctuations within the structure.

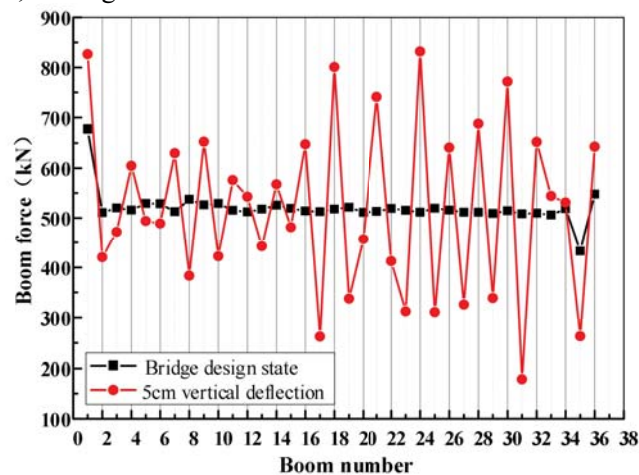


Figure 5. Comparison of cable force under 5cm deflection and no deflection.

4. Sensitivity Analysis of Target Cable Force Value

4.1. Target Deflection Amplitude

In order to analyze the impact of deflection deformation on the target cable force, the corresponding target cable forces are calculated for different deflections of the bridge control line. This analysis aims to assess the sensitivity of various bridge deformations on the target cable force. The target cable forces were computed for no deflection, mid-span deflections of 10mm, 20mm, 30mm, 40mm, and 50mm, as well as a pre-arch deflection of 10mm. The results are presented in Figure 6.

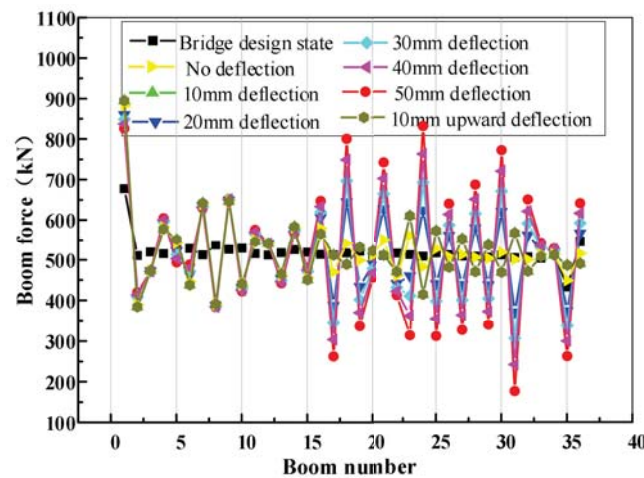


Figure 6. Target boom forces under different target deflection amplitude.

The cable adjustment amplitude of the bridge cable force is closely correlated with the target linear deflection displacement, as depicted in Figure 6. Moreover, the cable adjustment amplitude varies at different boom positions due to inconsistent influence coupling range in adjacent booms. Taking the influence range of internal forces of the first derrick as an example, analysis of the distribution characteristics of elements within the internal force influence matrix reveals that a change in unit force exerted by the first derrick results in varying amplitudes of internal forces for five directly affected adjacent booms. This indicates that four effective sling cables are impacted by the first rod, which is denoted as $n_1^{eff} = 4$. The different number of booms results in varying influence ranges, leading to distinct sensitivities of the cable force at different boom positions.

To facilitate comparative analysis, Formula (5) is employed to calculate the percentage change amplitude of cable force in the derrick relative to the bridge's state. The summarized cable adjustment amplitudes α for different booms are presented in Figure 7.

$$\alpha = (P_c/P_0 - 1) \times 100\% \quad (5)$$

The graph in Figure 7 illustrates that when the bridge's linear shape is transformed into a parabolic form, the amplitude of variation in the adjusted cable force of the sling decreases as the deflection displacement value decreases. Additionally, it is most noticeable that there is an evident amplitude variation in cable force from the quarter span to the mid-span segment.

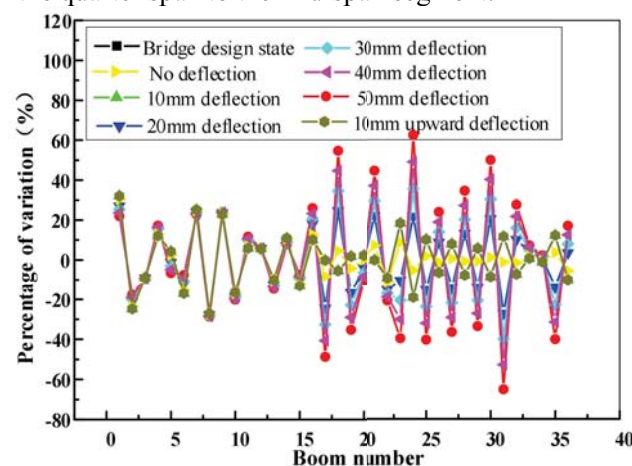


Figure 7. Variation amplitude of bridge cable force under different target deflection amplitude.

4.2. Target Line Shape

To investigate the distribution law of cable force for different linear shapes, the parabolic and sine curves to calculate the target cable force at mid-span displacements of 10mm and -10mm are employed. The results of these calculations are presented in Figure 8. It can be observed from Figure 8 that there is minimal variation in the target cable force values with changes in line shape, with the largest difference occurring at derrick No. 2, amounting to only 0.64%. Consequently, it can be concluded that the target cable force during bridge formation is not significantly affected by variations in line shape, rendering any influence negligible.

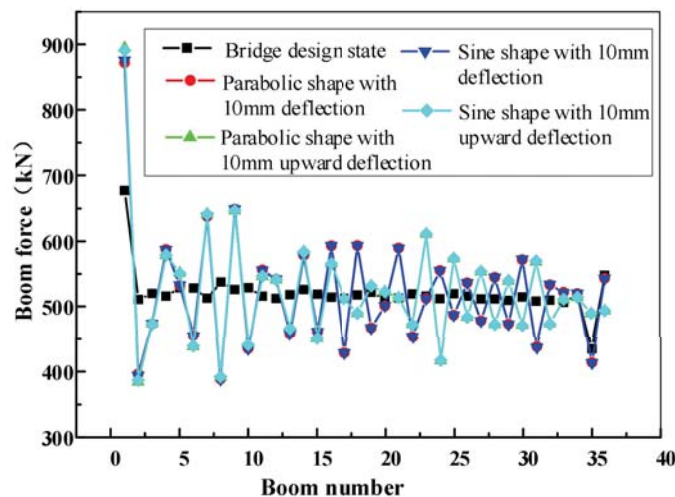


Figure 8. Analysis of the influence of line shape difference on the bridge cable force value.

5. Conclusion

The target cable force of the suspension bridge replacement sling is calculated based on the double matrix analysis model. The influence of bridge alignment and deflection displacement on the target cable force is analysed using the Bridge A suspension replacement project as a case study, leading to the following conclusions:

- 1) The cable forces between adjacent cable rods in the target state will exhibit significant nonlinear fluctuations when the predetermined target alignment is deemed unreasonable.
- 2) The adjustment amplitude of the bridge cable force is closely correlated with the target linear deflection displacement, and the cable adjustment amplitude varies at different boom positions due to inconsistent coupling ranges of influence in adjacent booms.
- 3) The target cable force of the completed bridge is insensitive to its shape, and the differences in shape can be disregarded when calculating the target cable force.
- 4) The change amplitude of the cable force decreases as the deflection target displacement decreases, and the change of cable force from quarter span to mid-span segment is the most obvious.

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Conflicts of Interest

There is no conflict of interest.